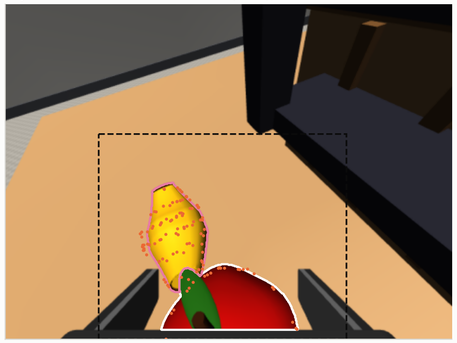
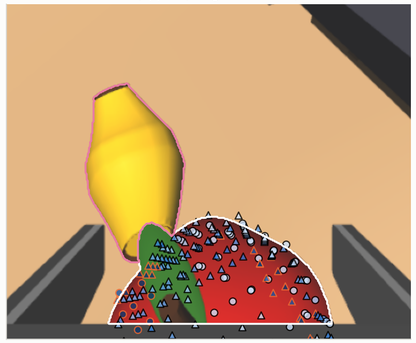
T96 / T97 fixed-input replay: hand layer at one grasp-window request (same input, four hand-layer rules)

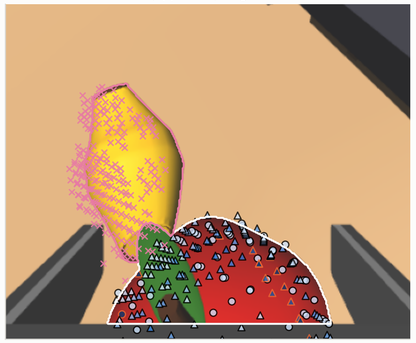
REG ep211: apple touching a banana
request 23, palm-apple 13 mm, grasping-hand wrist camera
Layout: horizontal slice z = 0.852 m (base), seen from above



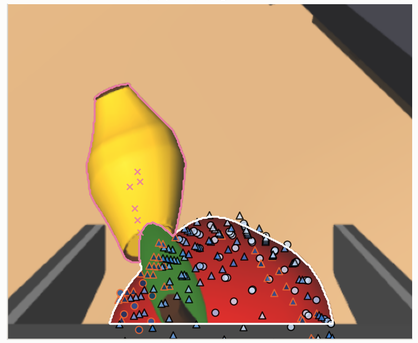
No feature
(surface extraction only)
remaining 74/382 = 0.194 (visible 7, hidden 67)
freed vs no feature: 0 (apple 0, neighbour fruit 0, table 0, crate 0)



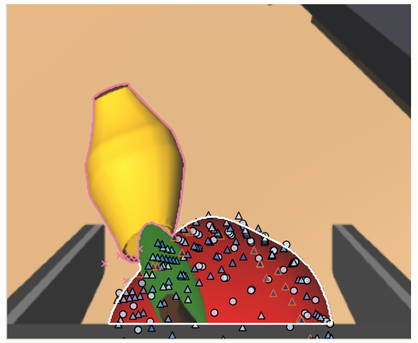
P0m hand layer
(hull fill + contact allowance 10 mm)
remaining 45/382 = 0.118 (visible 1, hidden 44)
freed vs no feature: 355 (apple 109, neighbour fruit 236, table 0, crate 0)



Target solid, d 0 mm
(T96, tau 5 mm)
remaining 103/382 = 0.270 (visible 7, hidden 96)
freed vs no feature: 13 (apple 7, neighbour fruit 6, table 0, crate 0)



P0m hand layer + crease cut
+ protrusion re-attach + hand freeze 80 mm (T97)
remaining 47/382 = 0.123 (visible 0, hidden 47)
freed vs no feature: 144 (apple 135, neighbour fruit 7, table 0, crate 0)

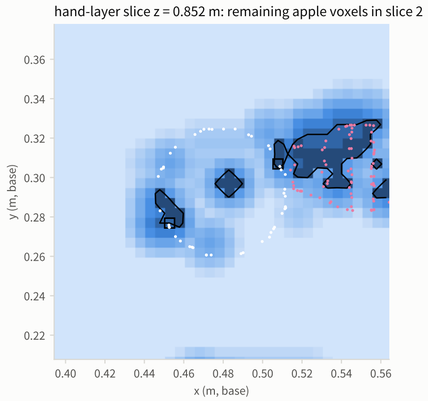


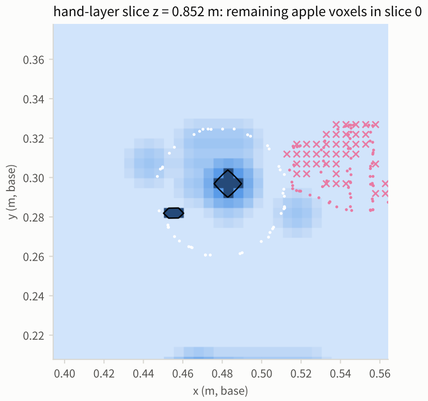
How to read (both rows)

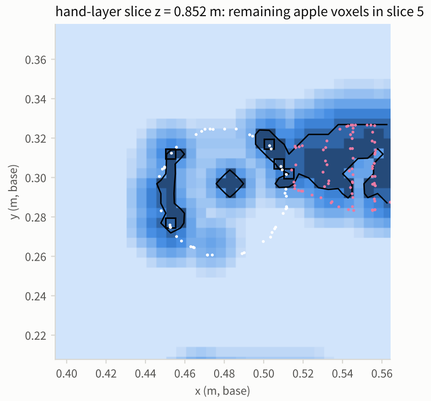
Top row: wrist-camera render (dashed box at left = zoomed area) with band-excluded GT apple surface voxels projected, coloured by the hand-layer value d (mm). Orange edge = d <= 0 (still an obstacle = remaining target voxel). Circle = visible in this camera, triangle = hidden. Pink x = voxel freed vs no-feature whose GT class is a neighbour fruit (table / crate in other colours).

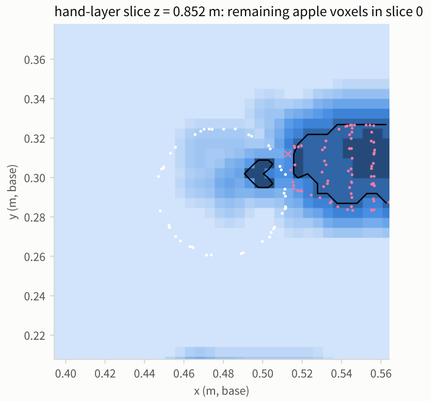
Bottom row: hand-layer slice at the plane shown left (top view, +x right, +y up, base frame). Black line = d = 0. White dots = GT apple surface in the slice; pink dots = GT neighbour fruit. Black squares = remaining apple voxels in the slice.

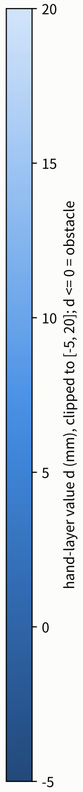
White / pink lines in the camera = GT apple / neighbour-fruit outline (MuJoCo segmentation).



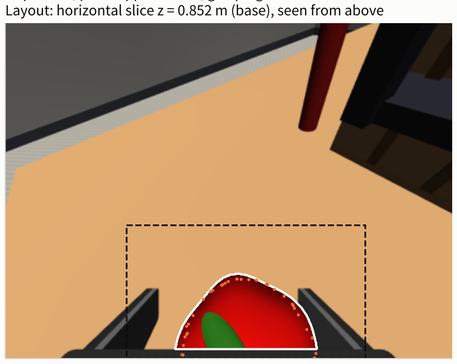




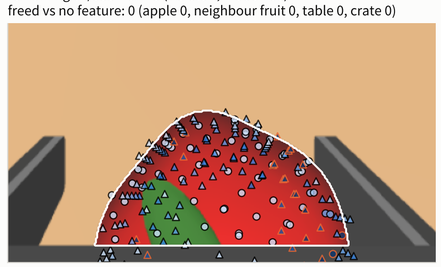




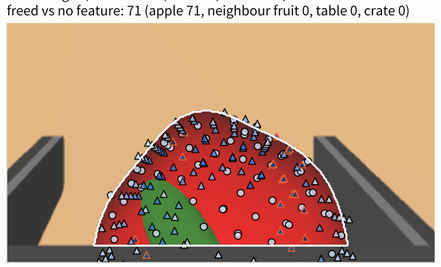
HURDLE ep1345: apple separated from neighbours
request 20, palm-apple 14 mm, grasping-hand wrist camera
Layout: horizontal slice z = 0.852 m (base), seen from above



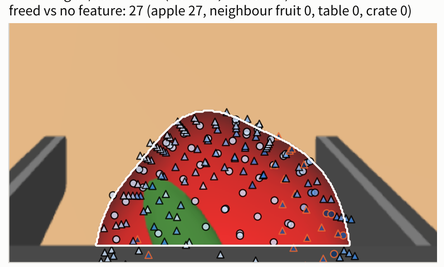
No feature
(surface extraction only)
remaining 62/358 = 0.173 (visible 2, hidden 60)
freed vs no feature: 0 (apple 0, neighbour fruit 0, table 0, crate 0)



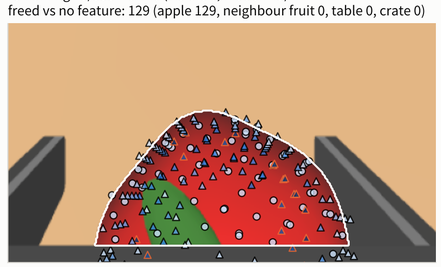
P0m hand layer
(hull fill + contact allowance 10 mm)
remaining 45/358 = 0.126 (visible 0, hidden 45)
freed vs no feature: 71 (apple 71, neighbour fruit 0, table 0, crate 0)



Target solid, d 0 mm
(T96, tau 5 mm)
remaining 67/358 = 0.187 (visible 4, hidden 63)
freed vs no feature: 27 (apple 27, neighbour fruit 0, table 0, crate 0)



P0m hand layer + crease cut
+ protrusion re-attach + hand freeze 80 mm (T97)
remaining 43/358 = 0.120 (visible 0, hidden 43)
freed vs no feature: 129 (apple 129, neighbour fruit 0, table 0, crate 0)

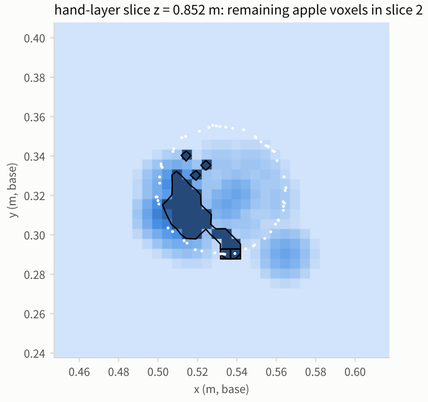


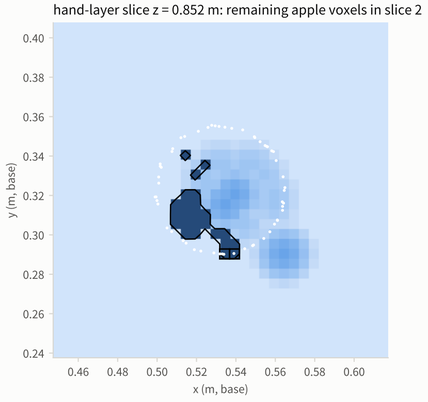
How to read (both rows)

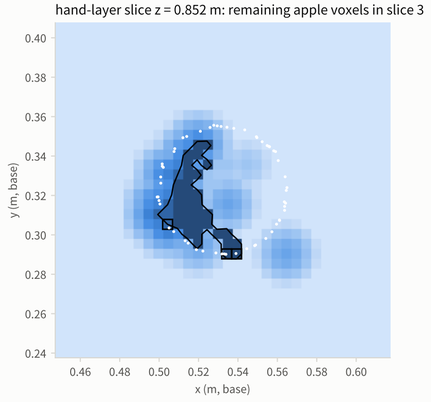
Top row: wrist-camera render (dashed box at left = zoomed area) with band-excluded GT apple surface voxels projected, coloured by the hand-layer value d (mm). Orange edge = d <= 0 (still an obstacle = remaining target voxel). Circle = visible in this camera, triangle = hidden. Pink x = voxel freed vs no-feature whose GT class is a neighbour fruit (table / crate in other colours).

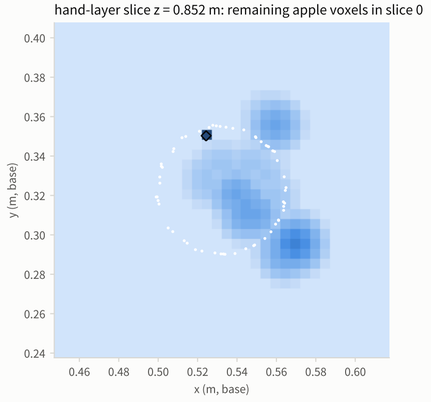
Bottom row: hand-layer slice at the plane shown left (top view, +x right, +y up, base frame). Black line = d = 0. White dots = GT apple surface in the slice; pink dots = GT neighbour fruit. Black squares = remaining apple voxels in the slice.

White / pink lines in the camera = GT apple / neighbour-fruit outline (MuJoCo segmentation).









remaining apple voxel, visible (d <= 0)
remaining apple voxel, hidden (d <= 0)
apple voxel not an obstacle (d > 0)
freed voxel, GT neighbour fruit
freed voxel, GT table
freed voxel, GT crate
GT apple outline (drawn white)
GT neighbour fruit outline
slice plane on GT surfaces (layout)
remaining apple voxel in slice